

AGWU SUNDAY EZE

Graph Data Science & GraphRAG AI Engineer

Port Harcourt, Nigeria | contact@adjacency.xyz

Portfolio: <https://adjacency.xyz> | **LinkedIn:** <https://www.linkedin.com/in/sonnyhoney> | **GitHub:** <https://github.com/sonnyhoney>

PROFESSIONAL SUMMARY

Neo4j Certified Graph Data Science Engineer specializing in constructing enterprise Knowledge Graphs, designing high-throughput Cypher schemas, and implementing hybrid GraphRAG architectures with Large Language Models (LLMs). Proven track record in parsing unstructured multi-format data (PDF, DOCX, CSV), modeling connected property graphs, and executing graph machine learning algorithms (PageRank, Louvain, FastRP) to eliminate hallucinations and deliver deterministic, auditable business intelligence.

TECHNICAL SKILLS

- **Graph Technologies & Databases:** Neo4j (Aura Cloud, Desktop, Enterprise), Cypher Query Language, Graph Data Science (GDS) Library, APOC Procedures, Property Graph Modeling, Schema Design, Index Optimization.
- **Graph Machine Learning & Analytics:** Node Embeddings (FastRP, Node2Vec), Community Detection (Louvain, WCC), Centrality Algorithms (PageRank, Betweenness), Link Prediction, NetworkX, PyVis.
- **Artificial Intelligence & LLMs:** Universal GraphRAG, Knowledge Graphs, Google Gemini Flash (google-genai), Multimodal OCR, Prompt Engineering, Grounded Retrieval, Hallucination Prevention.
- **Languages & Backend:** Python (OOP, Async), Pandas, NumPy, FastAPI, Streamlit, PyMuPDF, python-docx, Git, Bash Scripting, SQL.
- **Data Engineering & Cloud:** Multi-format ETL Pipelines, Text Chunking Strategies, REST APIs, Vercel, Docker, CI/CD Workflows.

CERTIFICATIONS

- **Neo4j Certified Professional** — Neo4j / GraphAcademy | 2026
Validated mastery in Cypher query tuning, sub-graph querying, and enterprise graph data modeling.
- **Neo4j Graph Data Science (GDS) Certified** — Neo4j / GraphAcademy | 2026
Specialized in graph algorithms, node embedding projections, and predictive graph machine learning pipelines.
- **Google IT Automation with Python Professional Certificate** — Google | 2025
Validated expertise in system automation, Python data pipelines, Git version control, and scripting.
- **Google Prompting Essentials** — Google | 2024
Advanced prompt design, structured output constraints, and LLM reasoning workflows.

PROFESSIONAL EXPERIENCE

Independent Graph AI & Data Science Consultant | Remote

March 2026 – Present

- Architect and deploy custom **GraphRAG pipelines** and **Knowledge Graphs** for enterprise clients, achieving **98%+** retrieval accuracy and eliminating hallucinations across **50k+** multi-hop document nodes.
- Engineer end-to-end ETL pipelines in Python converting unstructured PDFs, contracts, and tabular data into interconnected Neo4j property graph schemas.
- Optimize Cypher execution plans and index-backed traversals, reducing graph context retrieval latency to **sub-300ms** across dense multi-hop node paths.
- Develop interactive full-stack client dashboards using **Streamlit** and **PyVis**, accelerating stakeholder insight discovery by **45%** through real-time graph physics simulations.

Data Scientist | Benetoys Nigeria Limited — Kaduna / Remote

September 2023 – December 2025

- Designed and maintained end-to-end data extraction workflows in Python, processing over **1,000+** unstructured documents and inventory records per month.
- Implemented structured data models and relational/graph database schemas to streamline reporting and eliminate duplicate entity records by **25%+**.
- Collaborated with cross-functional engineering teams to integrate machine learning and LLM-assisted features, improving operational workflow throughput by **30%**.
- Authored comprehensive technical documentation and CI/CD deployment pipelines on GitHub, increasing unit test suite code coverage to **90%+**.

Data Analyst | Udi Homes — Port Harcourt, Nigeria

November 2021 – June 2023

- Extracted, cleaned, and transformed **100,000+** property and customer records using Python (Pandas, NumPy) and SQL for executive decision support, reducing data inconsistency errors by **40%**.
- Streamlined manual reporting processes using scheduled Python scripts, saving **10+** engineering hours weekly.
- Built interactive metric dashboards tracking **15+** operational KPIs, improving executive decision-making speed by **25%**.

FEATURED ENGINEERING PROJECTS

Enterprise GraphRAG Engine & Trade Intelligence Platform

Live Demo: <https://adjacency.xyz/graphrag> | GitHub: <https://github.com/sonnyhoney/Enterprise-GraphRAG-Engine>

- **Architecture:** Neo4j Aura Cloud GDS, Google Gemini Flash, Streamlit, PyVis, PyMuPDF, Python.
- Built an enterprise-grade GraphRAG platform that extracts structured knowledge triples from multi-format files (PDF, DOCX, CSV, TXT, scanned OCR) into **Neo4j Aura Cloud**.
- Engineered a **2-Hop topological Cypher traversal engine** (`MATCH (a)-[r]->(b)-[r2]->(c)`) that retrieves grounded facts to augment LLM prompts with **zero hallucinations**.
- Integrated multi-page sliding chunking and dynamic PyVis physics visualization, allowing users to inspect subgraphs and audit trails with verifiable source citations.
- Benchmarked the platform on the **AfCFTA cross-border regulatory framework** for programmatic tariff and trade compliance validation.

Financial Fraud Ring & Circular Transaction Detector

GitHub: <https://github.com/sonnyhoney>

- **Architecture:** Neo4j Graph Data Science (GDS), Cypher, Louvain, Leiden, Weakly Connected Components (WCC), FastRP, Python.
- **Projected** large-scale transaction subgraphs into Neo4j GDS in-memory graph catalog to track synthetic identities and complex circular money-laundering patterns across **50,000+** simulated financial transactions.
- **Benchmarked and deployed** Louvain and Leiden community detection algorithms alongside WCC to isolate highly cohesive fraud clusters, eliminating disconnected sub-graph noise and improving anomaly detection accuracy by **35%** over baseline tabular heuristics

EDUCATION

PGDip in Statistics

University of Port Harcourt | 2025